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Tenkasi District
Common Half Yearly Examination - December 2025

Standard 12

Time Allowed: 3.00 Hours **BIOLOGY** Maximum Marks: 70

BIO-BOTANY

Marks: 35

I. Choose the best answer: **8×1=8**

1) The first cell of male gametophyte in Angiosperm is _____
   a) Microspore b) Megaspore
   c) Nucleus d) Primary Endosperm Nucleus
2) How many different kinds of gametes willbe produced by a plant having the genotype AABbCC?
   a) Three b) Four c) Nine d) Two
3) Changing the codon AGC to AGA represents _____
   a) Missense Mutation b) Non sense mutation
   c) Frameshift Mutation d) Deletion Mutation
4) ECORI cleaves DNA at _____
   a) AGGGTT b) GTAATC c) GAATTC d) TATAGC
5) The Prevention of Large Scale loss of biological interity _____
   a) Biopatent b) Bio Ethics c) Bio-Safety d) Bio-fuel
6) Father of Modern Ecology _____
   a) Alexander Von Humbolt b) Eugene P.Odum
   c) R.Misra d) Reiter
7) Van Mahostav day celebrated on _____
   a) September 16 b) July 07 c) March 21 d) April 22
8) Botanical name of Black gram _____
   a) Vigna Mungo b) Cajanus Cajan c) Vigna radiata d) Oryza Sativa

II. Answer any 4 questions: **4×2=8**

9) Any two uses of Black Pepper.
10) What is Cantharophily?
11) Define Backcross.
12) What is meant by shuttle vector?
13) What is phytoremediation?
14) Define 10% law.

III. Answer any three questions and question number 19 is compulsory: 3×3=9

15) Differentiate primary and secondary introduction.
16) What is meant by cytoplasmic inheritance?
17) Advantages any Three Herbicide Tolerant Crops?
18) Write short notes on Pollen Kitt.
19) Tabulates plant indicators for pollution.

IV. Answer all questions: **2×5=10**

20) Explain the following:
   a) What are psychoactive drugs? Add note on Marijuana and Opium. -3
   b) Economic importance of Rice -2

(OR)

Application of Bio technology.

21) Bring out the inheritance of Chloroplasts gene with example. -5

(OR)

Important characters of Anemophilous flowers. -5

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BIO-ZOOLOGY

Marks: 35

I. Choose the best answer: **8×1=8**

1) SMOG is derived from _____
   a) smoke b) FOG c) Both A & B d) Only A
2) Which one of the following are at high risk extinction due to habitat destruction _____
   a) Mammals b) Birds c) Amphibians d) Echinoderm
3) The First Clinical gene therapy was done for the treatment of _____
   a) AIDS b) Cancer c) Cystic fibrosis d) SCID
4) _____ is the Major Producer of Ethanol
   a) Bacteria b) Fungi c) Yeast d) Virus
5) _____ is the Secondary Lymphoid Organ
   a) Liver b) Tonsil c) Thymus d) Spleen
6) The Wings of Birds and butterflies is an example of _____
   a) Adaptive Radiation b) Analogous Structures
   a) Divergent Evolution d) Natural Selection
7) Embryo implantation is occurs at _____
   a) Vagina b) Fallopian tube c) Uterus d) Peritoneal Cavity
8) _____ is hormone releasing IUD
   a) CUT-380A b) CU7 c) Multiload 375 d) Progestasert

II. Answer any four questions: **4×2=8**

9) Define Thelytoky and Amphitoky.
10) Expand the following a) FSH b) LH c) hCG d) hPL
11) What is Lyonisation?
12) Why tRNA is called an Adapter molecules?
13) What is Oenology?
14) Differenciate Stenotherms and Eurytherms.

III. Answer any three questions. Q.No. 19 is compulsory: **3×3=9**

15) Explain Von't Hoff's rule.
16) What is Acclimatisation?
17) Mention the main objections of Darwinism.
18) Differentiate Foeticide and Infanticide.
19) Identify the Diagram and Write the labelled parts:



IV. Answer all questions: **2×5=10**

20) Explain the Parturition and Lactation. -5

(OR)

Explain the Marriage between colour blind Man and Normal Women. -5

21) Explain the following:
   a] Salient features of mutation theory -3
   b] Somatic cell gene therapy and Germ Line Gene Therapy -2

(OR)

Analyse of two species population interaction. -5

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